

# Dissertation Research Proposal

Jahangeer Khan · MSc Artificial Intelligence · Sheffield Hallam University · April 2026

|                       |                                           |
|-----------------------|-------------------------------------------|
| Student               | Jahangeer Khan                            |
| Programme             | MSc Artificial Intelligence               |
| Supervisor            | Prof. Jing Wang (Approved ✓)              |
| Module Leader         | Alice Arnold                              |
| Module                | Research Skills for Computing (55-710248) |
| Proposal Deadline     | 12th May 2026                             |
| Dissertation Deadline | September 2026                            |

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Dissertation Title & Research Question

*Detecting Concept Drift in Non-IID Federated Learning Environments*

Research Question:

*"Can cosine similarity detect concept drift in a Non-IID Federated Learning environment?"*

|             |                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aim         | To investigate whether temporal cosine similarity of a client's own gradient history can reliably distinguish a drifting client from a Non-IID client in Federated Learning. |
| Domain      | IoT (Internet of Things) — sensor networks and smart devices                                                                                                                 |
| Application | Detecting sensor degradation and environmental drift in distributed IoT FL systems                                                                                           |
| Approach    | Double sliding window cosine similarity — short window (5 rounds) for sudden drift, long window (50 rounds) for gradual drift                                                |

|                  |                                                                                  |
|------------------|----------------------------------------------------------------------------------|
| <b>Baselines</b> | FedAvg (standard FL) vs CDA-FedAvg (existing drift detection) vs Proposed Method |
| <b>Datasets</b>  | UCI Human Activity Recognition / Gas Sensor Array (IoT) + CIFAR-10 (benchmark)   |
| <b>Tools</b>     | Python, Flower (flwr), PyTorch, NumPy, Matplotlib, Scikit-learn                  |

## 2 Problem Statement

Federated Learning (FL) enables multiple devices or organisations to collaboratively train a shared AI model without sharing raw data. Each client trains locally and only sends model weights to a central server — preserving privacy while benefiting from collective learning.

In real-world FL deployments — particularly in IoT environments — two critical challenges arise simultaneously:

| Challenge     | What It Means                                                                                                                              | Server Action             |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Non-IID Data  | Each IoT sensor always has different data distributions — e.g. a factory sensor vs a home sensor. This is permanent and expected.          | Ignore — this is normal   |
| Concept Drift | A sensor's data suddenly or gradually changes over time — e.g. sensor degradation, seasonal change, equipment failure. This is unexpected. | Act — reduce contribution |

The core problem: The central server currently has NO memory of each client's historical behaviour. It cannot tell whether an unusual update is because the client was always different (Non-IID — ignore) or because something has changed (Concept Drift — act). Standard FedAvg treats both the same, causing model performance to degrade when drift occurs undetected.

## 3 Research Gap

A thorough review of existing FL literature reveals the following critical gap:

| Paper                             | What They Did ■                                 | What They Missed ■                                   |
|-----------------------------------|-------------------------------------------------|------------------------------------------------------|
| McMahan et al. (2017) — FedAvg    | Introduced standard FL — gold standard baseline | Assumes stationary IID data — ignores drift entirely |
| Casado et al. (2022) — CDA-FedAvg | Added concept drift detection to FL             | Only tested on IID data — Non-IID problem ignored    |

|                                             |                                                      |                                                           |
|---------------------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| Canonaco et al. (2021)<br>— Adaptive-FedAvg | Adjusts learning rate when drift detected            | Does not handle Non-IID distributions across clients      |
| Mahdi et al. (2025) —<br>Systematic Survey  | First comprehensive survey of FL under concept drift | Confirms: combined Non-IID + Drift is still unsolved      |
| FL-MalDrift (2025) —<br>Nature              | ADWIN-based drift detection in FL                    | Domain-specific, does not use gradient history comparison |

***GAP: No existing work uses a client's own gradient history with a double sliding window approach to distinguish Non-IID heterogeneity from both sudden AND gradual concept drift in FL. This is the novel contribution of this dissertation.***

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Proposed Solution — Double Sliding Window Cosine Similarity

This dissertation proposes a Double Sliding Window Cosine Similarity mechanism that gives the FL server memory — enabling it to detect both sudden and gradual concept drift by comparing each client's current gradients to its own historical gradient patterns.

| Component                    | Description                                                                                                                                                                                      | Detects                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Short Window (last 5 rounds) | Computes average cosine similarity of the most recent 5 training rounds. A sudden significant drop triggers immediate drift alert.                                                               | Sudden Drift (e.g. sensor failure) |
| Long Window (last 50 rounds) | Computes average cosine similarity over the last 50 rounds — the client's baseline behaviour. A consistent small gap between short and long window over 10+ rounds triggers gradual drift alert. | Gradual Drift (e.g. sensor ageing) |
| Adaptive Weighting           | When drift detected — server reduces that client's contribution weight from 100% to 30%. Weight gradually restores as client stabilises.                                                         | Both drift types                   |

**Research Contribution:** Finding the mathematical relationship between window size and the speed/type of drift that can be reliably detected — a specific, measurable, and novel experimental contribution.

| Condition      | Cosine Similarity Pattern                                         | Detection                                    |
|----------------|-------------------------------------------------------------------|----------------------------------------------|
| Normal Non-IID | Consistently low: 0.3, 0.3, 0.3, 0.3 — stable over time           | Ignore — normal                              |
| Sudden Drift   | Was high, suddenly drops: 0.8, 0.8, 0.8, 0.1 — one round change   | Alert immediately                            |
| Gradual Drift  | Slowly decreasing: 0.8, 0.75, 0.70, 0.65, 0.60 — over many rounds | Alert after 10 consecutive rounds of decline |

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Research Methodology & Design

|                   |                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------|
| Research Approach | Quantitative experimental — simulating FL scenarios and measuring performance metrics        |
| FL Framework      | Flower (flwr) — leading open-source FL framework, beginner-friendly, widely used in research |
| Clients           | 10 simulated IoT sensor clients                                                              |

|                    |                                                                               |
|--------------------|-------------------------------------------------------------------------------|
| Non-IID Simulation | Dirichlet distribution (alpha = 0.1 severe, 0.5 moderate, 1.0 mild)           |
| Sudden Drift       | Abrupt label shift at training round 30 — simulates sensor failure            |
| Gradual Drift      | Slow distribution shift over rounds 30-80 — simulates sensor degradation      |
| Primary Dataset    | UCI Human Activity Recognition / Gas Sensor Array (IoT domain)                |
| Benchmark Dataset  | CIFAR-10 (standard FL research benchmark for reproducibility)                 |
| Ethics             | No human participants — all datasets publicly available. UREC 1 form applies. |

Three Methods Compared:

| Method          | Description                                                                | Role                                             |
|-----------------|----------------------------------------------------------------------------|--------------------------------------------------|
| FedAvg          | Standard FL — averages all client weights with no drift awareness          | Baseline 1 — expected to fail under drift        |
| CDA-FedAvg      | Concept-Drift-Aware FedAvg (Casado et al. 2022) — existing drift detection | Baseline 2 — existing best solution to beat      |
| DAAW (Proposed) | Double sliding window cosine similarity + adaptive weighting               | Novel contribution — expected to outperform both |

Evaluation Metrics:

| Metric                         | What It Measures                                                          |
|--------------------------------|---------------------------------------------------------------------------|
| Global Accuracy                | Overall model accuracy across all clients — primary performance indicator |
| Recovery Speed                 | Number of rounds to recover accuracy after a drift event                  |
| Detection Precision/Recall     | How accurately the cosine similarity method identifies drift vs Non-IID   |
| Window Size vs Detection Speed | Mathematical relationship — the core novel experimental contribution      |
| Convergence Stability          | How stable the model remains under combined Non-IID + Drift conditions    |

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Literature Review — Key Papers

| Reference                                      | Contribution                                                           | Gap / Relevance to This Work                                                         |
|------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| McMahan et al. (2017)<br>FedAvg — AISTATS      | Original FL algorithm — FedAvg.<br>Gold standard baseline.             | Assumes stationary IID data. Your dissertation shows it fails under Non-IID + Drift. |
| Kairouz et al. (2021)<br>Open Problems in FL   | Comprehensive list of 65+ open FL research problems.                   | Formally confirms Non-IID + Concept Drift as an unsolved open problem.               |
| Casado et al. (2022)<br>CDA-FedAvg             | First drift-aware FL extension of FedAvg.                              | Ignores Non-IID setting. Your second baseline — DAAW should outperform it.           |
| Ganguly & Aggarwal (2023)<br>arXiv:2211.12578  | Online FL with non-stationary detection using FedAvg + FedOMD.         | Informs drift detection design. Does not use gradient history comparison.            |
| Mahdi et al. (2025)<br>MDPI Electronics 14(22) | First systematic survey of FL under concept drift (PRISMA guidelines). | Directly confirms research gap — combined problem still unsolved as of 2025.         |
| FL-MalDrift (2025)<br>Nature Sci. Reports      | ADWIN-based drift detection framework for FL in IoT/malware.           | Shows IoT + FL drift is active area. ADWIN vs cosine similarity comparison possible. |

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Project Timeline

| Period                     | Phase                         | Key Deliverables                                                                                                                                                      |
|----------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2026<br>Weeks 1-4      | Setup + Literature + Basic FL | Read 6 key papers, install Flower + PyTorch, basic FedAvg running, Non-IID simulated, Drift simulated. Submit proposal 12th May.                                      |
| June 2026<br>Weeks 5-8     | Cosine Similarity + Detection | Double window cosine similarity implemented, threshold found, adaptive weighting working, CDA-FedAvg baseline implemented. Lit review + methodology chapters written. |
| July 2026 Weeks 9-12       | Multi-Domain + Evaluation     | IoT dataset tested, Finance dataset tested, all 3 methods compared across 3 datasets, stakeholder evaluation collected. Results + discussion chapters written.        |
| August 2026<br>Weeks 13-16 | Final Writing + Submission    | All 5 chapters finalised, proofread, Jing review completed, GitHub code published. Dissertation submitted.                                                            |

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Ethics, Risks & Professional Issues

| Area         | Details                                                                                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethics Form  | UREC 1 — No human participants. All datasets (CIFAR-10, UCI HAR, Gas Sensor Array) are publicly available under open licenses. No personal data involved.                                              |
| Risk 1       | Cosine similarity may not fully distinguish Non-IID from Drift in all scenarios. Mitigation: Test multiple window sizes and threshold values, report all findings honestly including negative results. |
| Risk 2       | Time constraints over 4 months. Mitigation: Clear weekly milestones, regular supervisor meetings with Jing Wang, scope limited to simulation-based experiments.                                        |
| Risk 3       | Technical difficulties with FL framework. Mitigation: Using well-documented Flower framework with active community support and tutorials.                                                              |
| Legal        | CIFAR-10 licensed for research use (MIT). UCI datasets publicly available. All code published on GitHub under open licence for reproducibility.                                                        |
| Professional | Research follows SHU academic integrity guidelines. AI Transparency Scale (AITS) declaration included. All sources properly cited in APA format.                                                       |

## 9 Real-World Applicability & Beneficiaries

Although this research uses simulated environments, the proposed solution is directly applicable to real-world FL deployments. The domain-agnostic nature of the cosine similarity approach means the same method applies across industries by simply changing the dataset.

| Industry             | FL Use Case                                                      | How This Research Helps                                              |
|----------------------|------------------------------------------------------------------|----------------------------------------------------------------------|
| IoT / Smart Industry | Multiple sensors training shared anomaly detection model         | Detects sensor degradation drift before it corrupts the global model |
| Finance / Banking    | Multiple banks training shared fraud detection model             | Detects when fraud patterns shift — prevents missed detections       |
| Cybersecurity        | Multiple organisations training shared intrusion detection model | Detects when attack patterns evolve — keeps defences current         |
| Autonomous Vehicles  | Multiple vehicles training shared perception model               | Detects when road/weather conditions cause data drift                |

Evaluation Plan: Results will be shared with IoT and AI professionals through a structured evaluation questionnaire to gather real-world feedback on the solution's practical applicability. Expert validation from supervisor Prof. Jing Wang will also be documented.

- McMahan, B., Moore, E., Ramage, D., Hampson, S., & Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, PMLR.
- Kairouz, P., McMahan, H. B., et al. (2021). Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1-2), 1-210.
- Casado, F. E., Lema, D., Criado, M. F., Iglesias, R., Regueiro, C. V., & Barro, S. (2022). Concept drift detection and adaptation for federated and continual learning. *Multimedia Tools and Applications*.
- Ganguly, B., & Aggarwal, V. (2023). Online federated learning via non-stationary detection and adaptation amidst concept drift. *arXiv:2211.12578*.
- Mahdi, O. A., Pardede, E., Bevinakoppa, S., & Ali, N. (2025). Federated learning under concept drift: A systematic survey. *Electronics*, 14(22), 4480. MDPI.
- Canonaco, G., Bergamasco, A., Mongelluzzo, A., & Roveri, M. (2021). Adaptive federated learning in presence of concept drift. *International Joint Conference on Neural Networks (IJCNN)*.
- FL-MalDrift. (2025). FL-MalDrift: A federated learning framework for malware detection under local concept drift. *Nature Scientific Reports*.